

CHAPTER 4: Implementation

4.1 Introduction

This chapter describes the system's development. The Web Application for Carbon Footprint Analyzer is based on the previous chapter, that is chapter 3 of this report. The goal of this chapter is to show every technique and software program used in developing the system and its interface design. The Web Application for Carbon Footprint Analyzer will be developed using HTML, CSS, JavaScript for the frontend part, and for the backend part will be using PHP and MySQL.

This chapter will then go into further detail on the system hierarchy. The system hierarchy will consist of a few modules, each of which has distinct functions. Each module, which includes sample of interfaces, functions, and manual instructions, all will be thoroughly discussed later in this chapter.

4.2 Implementation Tools

4.2.1 Visual Studio Code



Figure 4.1 Visual Studio Code Logo

Visual Studio Code is a free, open-source code editor developed by Microsoft. Visual Studio Code is chosen as primary development environment as it comes with built-in IntelliSense that supports code completion, semantic analysis, navigation, and refactoring capabilities.

4.2.2 HTML, CSS and JavaScript

The three core web technologies used in this project are HTML, CSS, and JavaScript, each of which has a unique function in the development of the system. HTML is used to create the structure and layout of the web pages. This including the placement of elements such as forms, buttons, tables and navigation components. CSS is used to style and improve the visual appearance of the interface by controlling elements like colour schemes, typography, spacing, and responsive layout to ensure a consistent and user-friendly design. JavaScript is used to add interactives and dynamic features to the system. It manages user input validation, real-time updates, and the interaction between the interface and the main application logic. The three components that are necessary to create contemporary, aesthetically pleasing, and interactive online applications are HTML, CSS, and JavaScript.

4.2.3 Laragon



Figure 4.2 Laragon

Laragon is a fast, lightweight, and portable local development environment designed for web development on Windows. It was utilized in this project to provide a local server environment necessary for developing and testing the web-based system prior to deployment. Laragon integrates essential components such as Apache, MySQL, and PHP, allow the developer to run and manage the application locally without the need for an active internet connection or external hosting service.

4.3 Web Application for Carbon Footprint Analyzer Prototype

There are two roles in this web application, the user and administrator, which both has distinct login page. Each role has different interface and functionality displayed. They will be directed to respective page after login into the system. The Web Application for Carbon Footprint Analyzer will have NINE (9) modules which each of them serves different functions.

4.3.1 System Hierarchy

In software system hierarchy, a system is organized into multiple levels, start from the whole system to subsystems, modules and components where each level has its own responsibilities. This structured approach simplifies the process of design, development, testing and maintenance. Furthermore, this structure ensures that modifications in any levels will have minimal impact on overall system, making the system more flexible and scalable. The main goal of software architecture is to reduce complexity by dividing systems into smaller components that communicate through clear interfaces allowing multiple teams to work on different modules simultaneously and improve overall process, to make it faster and more efficient (Jaenudin & Widyarto, 2025).

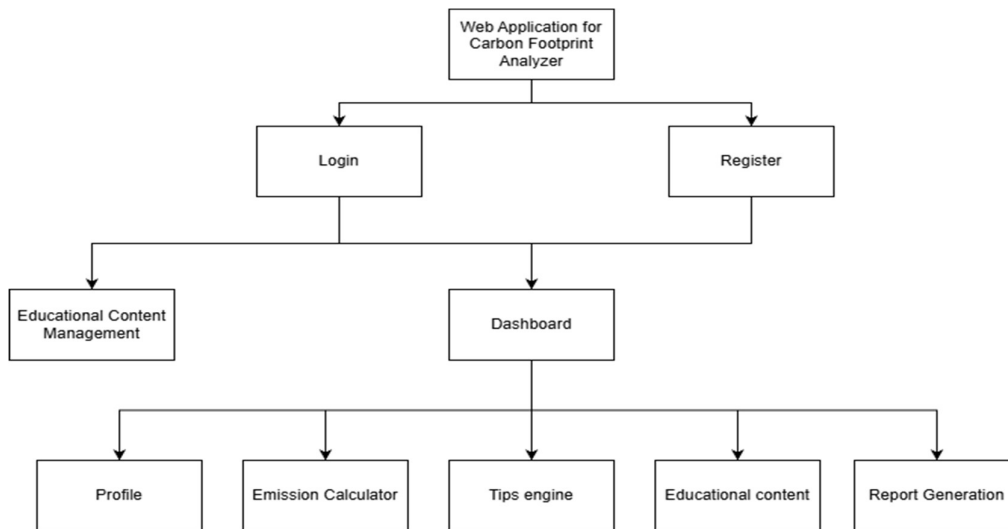


Figure 4.3 System Hierarchy for Web Application for Carbon Footprint Analyzer

There are NINE (9) modules in the system hierarchy for Web Application for Carbon Footprint Analyzer which are Login, Register, Dashboard, Profile, Emission Calculator, Tips Engine, Educational Content, Report Generation and Educational Content Management.

4.3.2 Login Module

The user is required to login into the system by entering their registered email address and password prior using other features. For a new user, they can register their account before continuing with the login process. After successfully login, the user will be directed to the user dashboard. Figure 4.4 show the user login for Web Application for Carbon Footprint Analyzer.

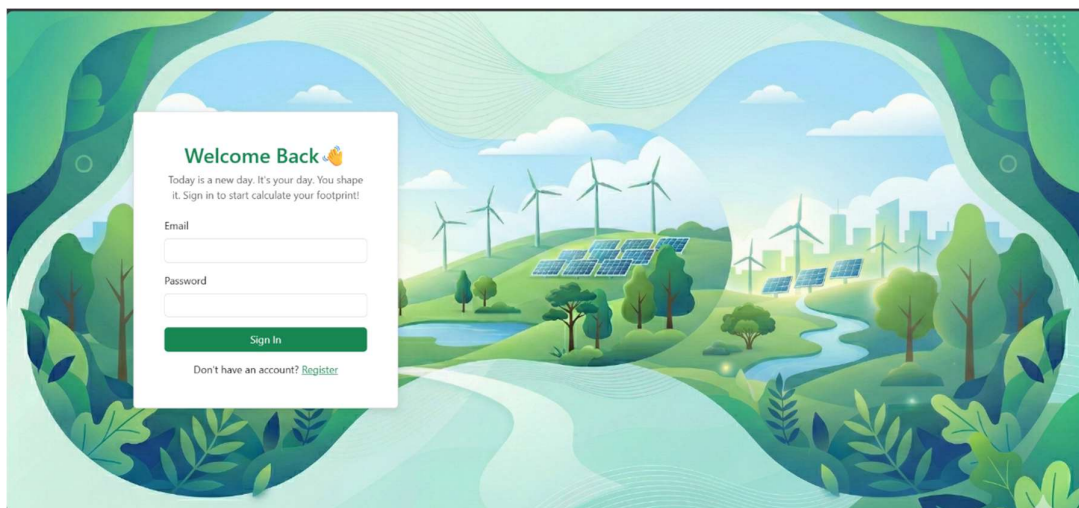


Figure 4.4 Login Page

4.3.3 Register Module

A user who does not have an account, they must create one. To register, the user must enter a few basic information such as name, email, password and select respective department. After successfully registered, the user should move to login page and can now login into their account to start using the system. Figure 4.5 shows the register page [for Web Application for Carbon Footprint Analyzer.](#)

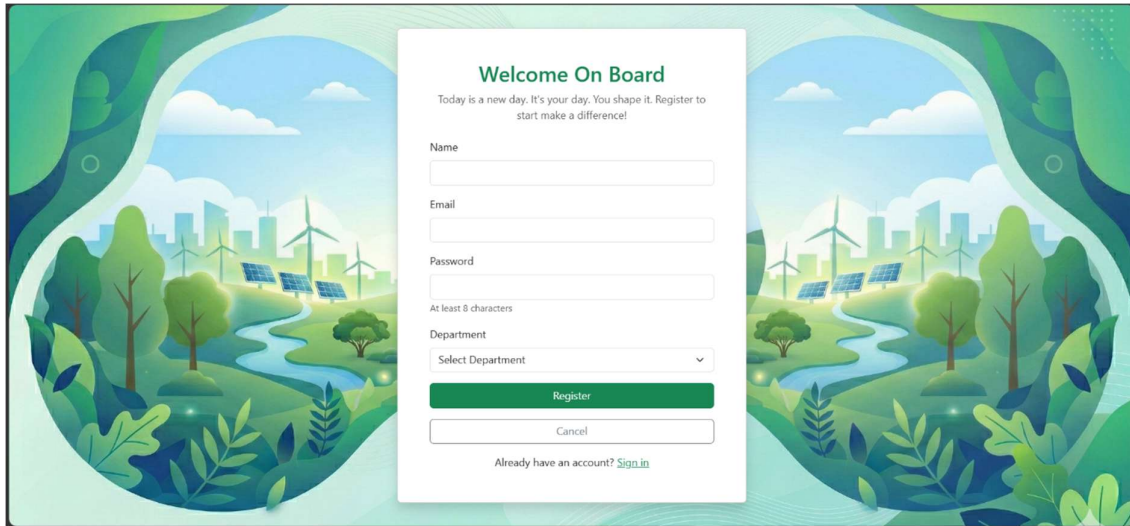


Figure 4.5 Register Page

4.3.4 Dashboard Module

The dashboard is the first page user will view after login where they will see welcoming message appear follow the current time. In the dashboard, user will be able to see their emissions data from previous input, like emissions for current and last month, latest entry, emissions trend, category breakdown and personalized tips. At the sidebar, there are few other functions such as profile, calculator, history and tips and education (As shown in Figure 4.6).

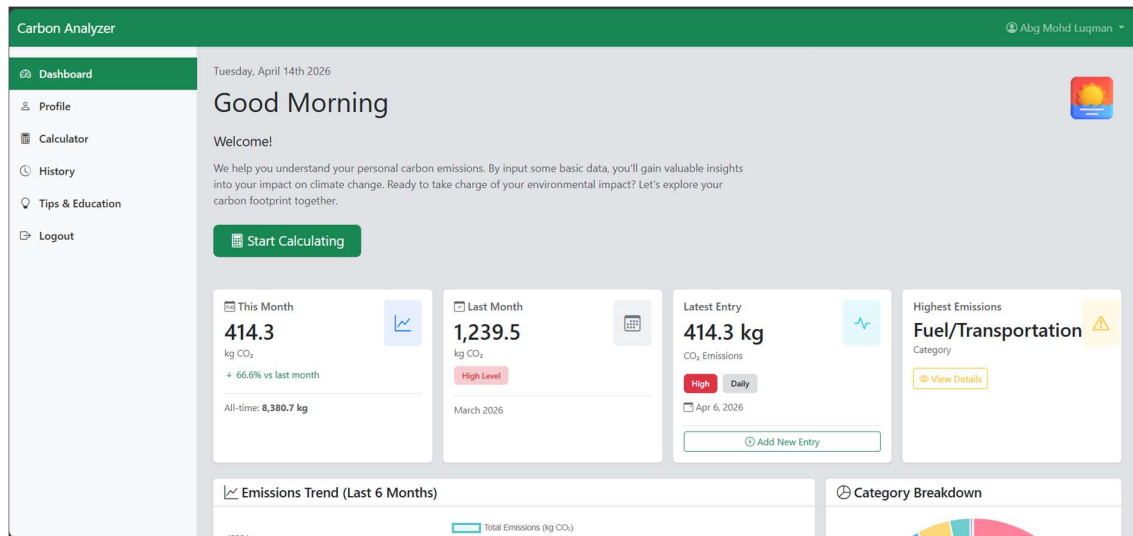


Figure 4.6 Dashboard Page

4.3.5 Profile Module

The Profile page displays user personal information and account details (As shown in figure 4.7). User may update their name, email address, department, and password if needed. This page also presents individual statistics, including total records submitted, total emissions recorded, and the dates of the first and latest data entries.

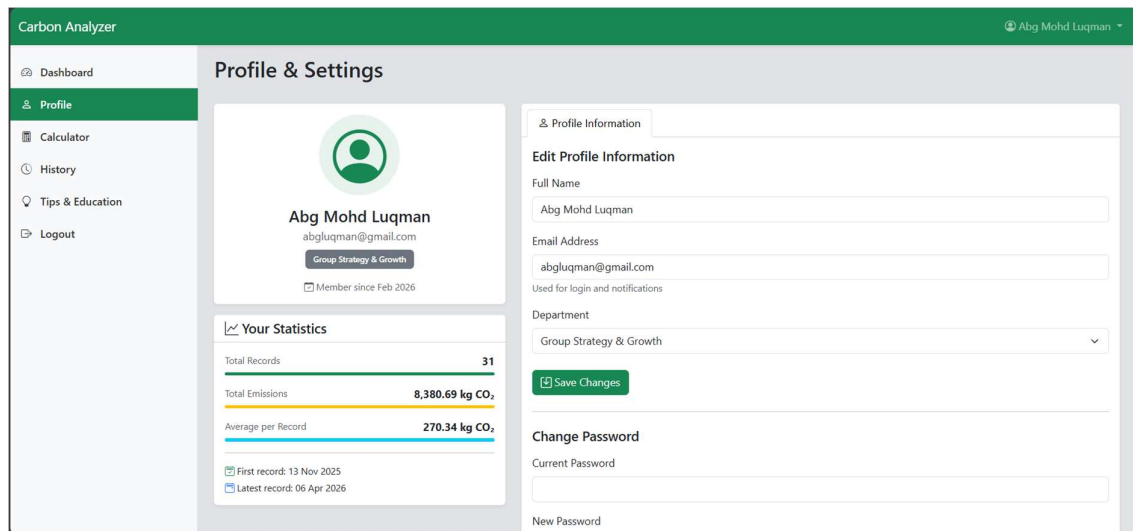
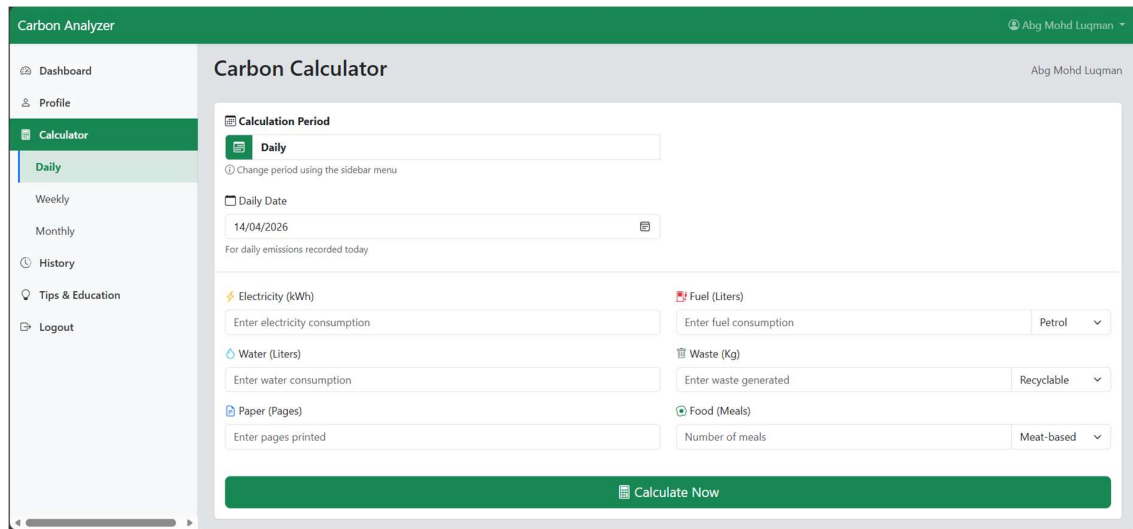


Figure 4.7 User Profile Page

4.3.6 Emission Calculator Module

The Emission Calculator page is where user input their consumption data for calculation. Users are required to provide basic consumption details, including electricity, water, fuel, paper, waste, and food (as shown in Figure 4.8). Data can be entered based on daily, weekly, or monthly consumption periods. Once submitted, the system calculate and displays the corresponding carbon emissions, helping user to better understand their environmental impact.



The screenshot shows the 'Carbon Calculator' interface within the 'Carbon Analyzer' application. The left sidebar contains navigation links: Dashboard, Profile, Calculator (selected), Daily (selected), Weekly, Monthly, History, Tips & Education, and Logout. The main content area is titled 'Carbon Calculator' and includes a user profile 'Abg Mohd Luqman'. The 'Calculation Period' is set to 'Daily'. Below this, there is a 'Daily Date' field showing '14/04/2026' and a note 'For daily emissions recorded today'. The form contains six input fields with corresponding icons and units: Electricity (kWh), Fuel (Liters), Water (Liters), Waste (Kg), Paper (Pages), and Food (Meals). Each field has a placeholder text 'Enter [consumption type]'. The 'Fuel (Liters)' field has a dropdown menu set to 'Petrol'. The 'Waste (Kg)' field has a dropdown menu set to 'Recyclable'. The 'Food (Meals)' field has a dropdown menu set to 'Meat-based'. At the bottom of the form is a green button labeled 'Calculate Now'.

Figure 4.8 Emission Calculator Page

4.3.7 Tips Engine Module

The Tips Engine is located at the bottom of the Dashboard page (as shown in Figure 4.9). This feature generates personalized recommendations to user based on their most recent consumption data entry. For instance, users with a high monthly consumption will receive different suggestions compared to those with a low monthly consumption, and vice versa. The system will determine the appropriate recommendations based on the consumption category that results the highest recorded value, ensures targeted and relevant guidance for each user.

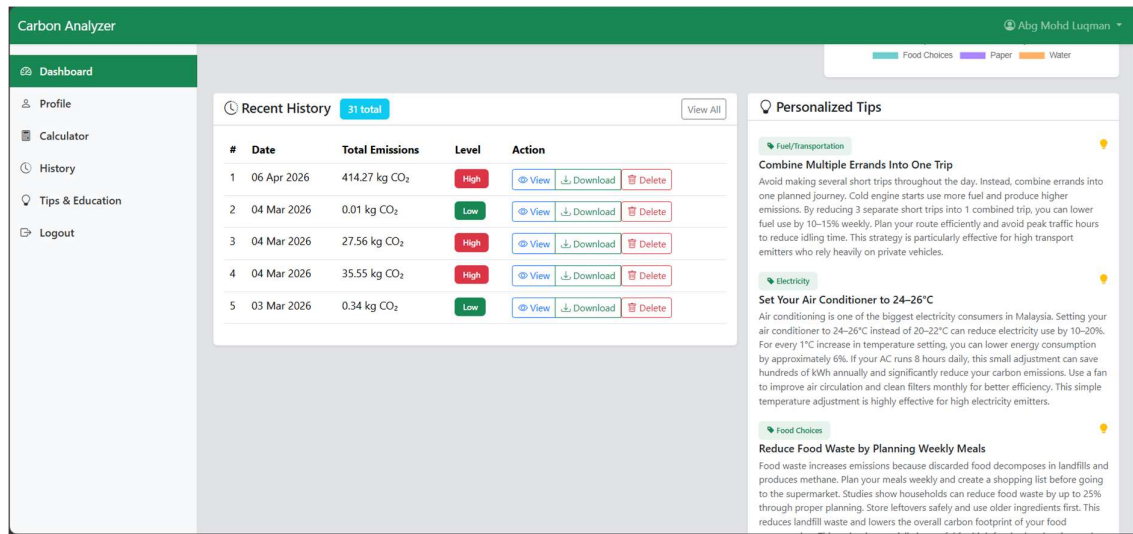


Figure 4.9 Dashboard Page (Tips Engine)

4.3.8 Educational Content Module

Educational content is accessible through the Tips & Education section (as shown in Figure 4.10). This section centralizes all educational content, including tips, articles, and guides across various categories and levels. User can apply filters to find content according to their preferences.

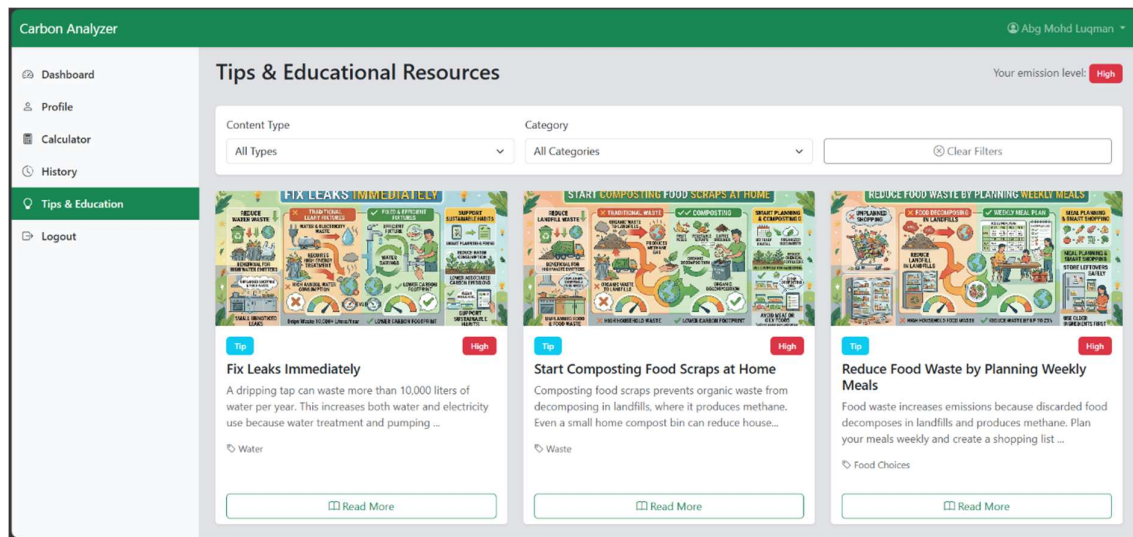


Figure 4.10 Educational Content Page

4.3.9 Report Generation Module

The History section is where user can access all their past emissions records in a simplified view (as shown in Figure 4.11). From here, user can download a report or delete

any previous entries. User can also filter their history by a preferred time period, choosing between daily, weekly, or monthly emissions.

Date	Total Emissions	Level	Details	Actions
06 Apr 2026 Monday	414.27 kg CO ₂	High	View	Download Delete
04 Mar 2026 Wednesday	35.55 kg CO ₂	High	View	Download Delete
04 Mar 2026 Wednesday	27.56 kg CO ₂	High	View	Download Delete
04 Mar 2026 Wednesday	0.01 kg CO ₂	Low	View	Download Delete
03 Mar 2026 Tuesday	42.57 kg CO ₂	High	View	Download Delete
03 Mar 2026 Tuesday	8.04 kg CO ₂	Medium	View	Download Delete
03 Mar 2026 Tuesday	124.62 kg CO ₂	High	View	Download Delete
03 Mar 2026 Tuesday	31.60 kg CO ₂	High	View	Download Delete

Figure 4.11 Emissions History

User are given the option to download their emissions report (as shown in Figure 4.12). The report compiles the user's carbon footprint result into a formal document suitable for official purposes, including personal basic information, current emissions data, and the last six months of emissions records for reference. Before downloading, user can preview the report and regenerate it if any changes are needed.

Carbon Footprint Report	
	Carbon Footprint Report Generated: 14 Apr 2026, 06:28 AM This report contains your detailed carbon emissions analysis, breakdown by category, historical trends, and personalized recommendations.
Download PDF	
Regenerate	
Report Preview	
Carbon Footprint Report Generated on 14 Apr 2026, 06:28 AM Personal Carbon Footprint Report Name: Abg Mohd Luqman Email: abgluqman@gmail.com Department: Group Strategy & Growth Report Date: 06 Apr 2026 Executive Summary Total Carbon Emissions 414.27 kg CO₂	

Figure 4.12 Report Generation Page

4.3.10 Educational Content Management Module

The admin page provides administrators with the ability to manage all educational content within the system. From this page, admins can view all uploaded educational content and make necessary changes by editing or deleting existing entries (as shown in Figure 4.13).

Admin
Carbon Footprint Analyzer

Dashboard

Users

Educational Content

View User Site

Logout

Administrator

Educational Content Management

Add New Content

Content Type: All Types | Emission Level: All Levels | Category: All Categories | Filter | Clear

ID	Title	Type	Category	Level	Author	Created	Actions
25	Understanding Your Personal Carbon Footprint	Tip	General	General	Administrator	02 Mar 2026	Edit Delete
24	Reduce, Reuse, Recycle — In That Order	Tip	General	General	Administrator	02 Mar 2026	Edit Delete
23	Small Daily Habits Create Long-Term Carbon Reductions	Tip	General	General	Administrator	02 Mar 2026	Edit Delete
22	Track Your Carbon Footprint Monthly	Tip	General	General	Administrator	02 Mar 2026	Edit Delete
21	Turn Off the Tap While Brushing Teeth	Tip	Water	Low	Administrator	02 Mar 2026	Edit Delete
20	Fix Leaks Immediately	Tip	Water	High	Administrator	02 Mar 2026	Edit Delete

Figure 4.13 Educational Content Management Page

When adding new educational content, administrators are required to provide several key details, including a title, description, content type, emissions level, and any relevant attachments such as images or videos. To ensure consistency, content guidelines are also provided on this page, keeping all administrators informed of the standards and requirements expected when uploading new material (As shown in Figure 4.14).

Admin
Carbon Footprint Analyzer

Dashboard

Users

Educational Content

View User Site

Logout

Administrator

Add Educational Content

Back to List

Title *

Enter content title

Description *

Enter content description

Provide detailed information, tips, or guidance

Content Type *

Select Type

Emission Level

General (All Levels)

Target specific emission levels for personalized tips

Category

General (All Categories)

Link to specific emission category or leave as general

Image (Optional)

Choose File | No file chosen

Max size: 16MB. Formats: JPG, PNG, GIF

Content Guidelines

Tips

- Short, actionable advice (50-150 words)
- Focus on one specific action
- Include measurable benefits when possible

Articles

- In-depth educational content (200-500 words)
- Explain concepts and research
- Provide context and examples

Videos

- Educational video content
- Multimedia presentations
- Step-by-step visual guides

Best Practices

- Use clear, simple language
- Be specific and actionable
- Include relevant statistics when available
- Keep user motivation in mind
- Review for accuracy before publishing

Figure 4.14 Educational Content Management (Add New Content)

4.4 Discussion

The implementation of Web Application for Carbon Footprint Analyzer shows that most core functionalities were successfully developed. The emissions calculator accurately calculates and process user input and generate reliable result. At the same time, the dashboard provides a clear visualization of user data.

4.5 Summary

References

Jaenudin, J., & Widyarto, S. (2025). *Architectural strategies for scalable, secure, and high-performance software systems*. Proceedings of Informatics Conference.
<https://ojs.journals.unisel.edu.my>